

Translated

From: e-maxx.ru

Heavy-light decomposition

Heavy-light decomposition is a fairly general technique that allows us to effectively solve many problems that come down to **queries on a tree**.

Description

Let there be a tree T of n vertices, with an arbitrary root.

The essence of this tree decomposition is to **split the tree into several paths** so that we can reach the root vertex from any v by traversing at most $\log n$ paths. In addition, none of these paths should intersect with another.

It is clear that if we find such a decomposition for any tree, it will allow us to reduce certain single queries of the form "calculate something on the path from a to b " to several queries of the type "calculate something on the segment $[l, r]$ of the k^{th} path".

Construction algorithm

We calculate for each vertex v the size of its subtree $s(v)$, i.e. the number of vertices in the subtree of the vertex v including itself.

Next, consider all the edges leading to the children of a vertex v . We call an edge **heavy** if it leads to a vertex c such that:

$$s(c) \geq \frac{s(v)}{2} \iff \text{edge } (v, c) \text{ is heavy}$$

All other edges are labeled **light**.

It is obvious that at most one heavy edge can emanate from one vertex downward, because otherwise the vertex v would have at least two children of size $\geq \frac{s(v)}{2}$, and therefore the size of subtree of v would be too big, $s(v) \geq 1 + 2 \cdot \frac{s(v)}{2} > s(v)$, which leads to a contradiction.

Now we will decompose the tree into disjoint paths. Consider all the vertices from which no heavy edges come down. We will go up from each such vertex until we reach the root of the tree or go through a light edge. As a result, we will get several paths which are made up of zero